

Product Design Specification Report

Group 8

1. Product Description Executive Summary (15%)

1.1 Team Name and Summary of Participants

- **Team Name:** “Take it E.Z.”
- **Team Members:** Gregory Sharma, Carmen Lin, Emma Weiss
- **Brief Summary of Participants and Roles:**
 - Gregory Sharma (GS): Mechanical engineering senior, experience with rapid prototyping and design. Within LoadStep, Greg outlined the product architecture and the interactions among the modules in the device. He also performed the mechanical analyses, outlined the value-based pricing, and completed the failure mode and effects analysis.
 - Carmen Lin (CL): Mechanical engineering junior with minors in business and engineering management, experience in rapid prototyping, CAD, fabrication, and business strategy. Within LoadStep, Carmen led customer interviews and conjoint analysis, using user insights to guide design decisions. She was also responsible for financial modeling and environmental impact assessment.
 - Emma Weiss (EW): Biomedical engineering senior & Mechanical engineering MEng student. Experienced with rapid prototyping, CAD design, design for manufacture and assembly, and needs finding. Within LoadStep, Emma primarily worked on developing the house-of-quality matrices, mapping customer solutions to engineering and part characteristics. She also was responsible for the bass model forecast, safety regulations, and IP concerns.

1.2 Product Name:

Product Name: The LoadStep

Product Slogan: Recover Through Motion

1.3 Problem you are solving

Athletes, either professional or recreational, do not utilize all of the time available to them when recovering from an ankle sprain. Our goal is to produce a product that allows athletes to strengthen damaged ligaments more directly, safely, and more frequently, thereby increasing their recovery time and post-injury performance.

1.4 Basic Functions of the product to solve the problem

LoadStep contains a lightweight shin “splint” designed to carry the load of unwanted inversions or pressure on damaged joints. This helps to prevent further unwanted strain. It also has adjustable

loading bands that work against ankle inversion and strengthen the ankle, similar to the strength training usually done in physical therapy. By integrating this technology directly into an ankle brace, the LoadStep allows users to strengthen their ankle while proceeding with their day-to-day life.

1.5 Special features of the product beyond the core problem solution

Beyond the core solution, the LoadStep includes compressive straps and comfortable padding to encourage the user to push through the discomfort of working through the damaged tissue. A water-absorbing internal layer achieves a similar effect, providing evaporative cooling and easing discomfort. A protective outer shell helps protect the ankle from unwanted impact and overextension.

1.6 Picture of highest level product refinement (CAD or physical)

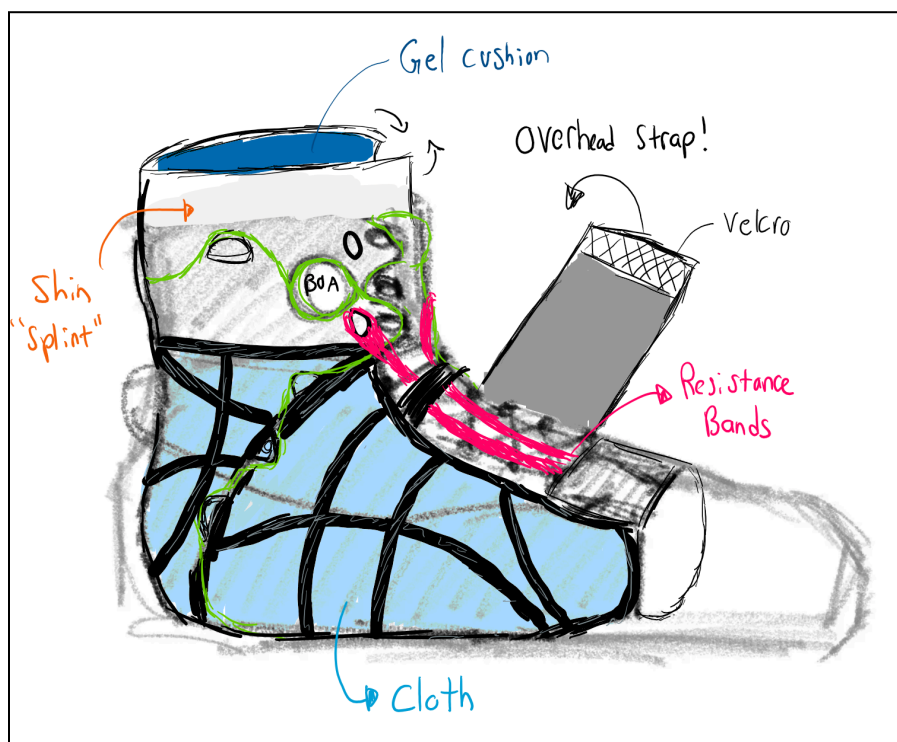


Figure 1: Final 2D Sketch of Product

2. Physical Description (33%)

2.1 House of Quality Level 1 (EW)

Our group created a House of Quality (HOQ) Level #1 matrix to link customer requirements to engineering characteristics (Table 1). Each customer requirement was assigned a weighting factor, with 5 as the most important and 1 as the least important. We then defined our engineering characteristics, indicating both the direction of improvement and the units for all parameters. The matrix is then filled out, assigning 9 when the engineering characteristic highly corresponds to the customer requirement, 3

when there is some relationship, 1 when there is little relationship, and 0 when there is no relationship. To calculate the raw score, we multiplied the matrix values by the corresponding weighting factors and summed the results for each engineering characteristic. We then converted the values into relative weights and ranked them.

Table 1: House of Quality #1 Matrix

Engineering Characteristics							
Improvement Direction		↑	↑	↑	↑	↓	↑
Units		Pa	N·m / °	Pa	s	g	N·m
Customer Requirements	Weight Factor	EC1: Brace Security	EC2: ROM Control	EC3: Compression Level	EC4: Cooling Level	EC5: Mass	EC6: Rehabilitation Loading
CR1: Comfortable	3	9	3	9	9	0	3
CR2: Cost-friendly	2	0	0	3	3	9	1
CR3: Ankle protection	4	9	3	3	0	3	3
CR4: Cold therapy	2	0	0	3	9	0	0
CR5: Installation Time	2	3	1	3	3	3	1
CR6: Light weight	4	3	3	1	3	9	0
CR7: Breatheable	1	3	0	3	3	3	0
CR8: Recovery Speed	5	3	9	1	1	0	9
Raw Score		99	80	69	77	75	70
Relative Weight		21.06%	17.02%	14.68%	16.38%	15.96%	14.89%
Rank Order		1	2	6	3	4	5

From this analysis, we learned that range of motion (ROM) control (EC2), cooling level (EC4), and brace security (EC1) are the three most important engineering characteristics to consider in our design.

2.2 Describe the List of Requirements from the Customer (EW/CL)

- **List of Customer Requirements (CRs)**

- Customer requirements are found in Table 1 and are listed here:
 - i. **CR1:** Comfortable
 - ii. **CR2:** Cost-friendly
 - iii. **CR3:** Ankle Protection
 - iv. **CR4:** Cold Therapy
 - v. **CR5:** Installation Time
 - vi. **CR6:** Light Weight
 - vii. **CR7:** Breatheable
 - viii. **CR8:** Recovery Speed

Customer Interviews

Fifteen interviews were conducted with student athletes, specifically recreational volleyball players, to better understand their experiences with ankle injuries and recovery. Several consistent themes emerged. Comfort was a major concern, as one participant stated, “I’m fine for the first bit, then it starts digging in, and I end up taking it off,” indicating that many current solutions fail during extended use.¹ Ankle protection was also important. As one participant noted, “Many of the current braces assume ankle immobilization, but it’s not always true.”¹ Some participants also mentioned that excess movement or poor fit can create instability and increase the risk of further injury. Recovery speed was another key factor, as athletes reported missing practices and games and expressed a strong preference for returning to play as quickly as possible. Cost was also a concern, particularly among college students with limited budgets. Finally, controlled loading emerged as an important need, as many athletes already follow rehabilitation exercises and seek a product that supports gradual, structured recovery.

2.3 Conjoint Analysis (CL)

Conjoint Test 1 (Initial)

Attribute	Levels
Price	\$40–65, \$65–90, \$90+
Comfort	Minimal, Cushioned, Gel
Loading	None, Constant, Adjustable

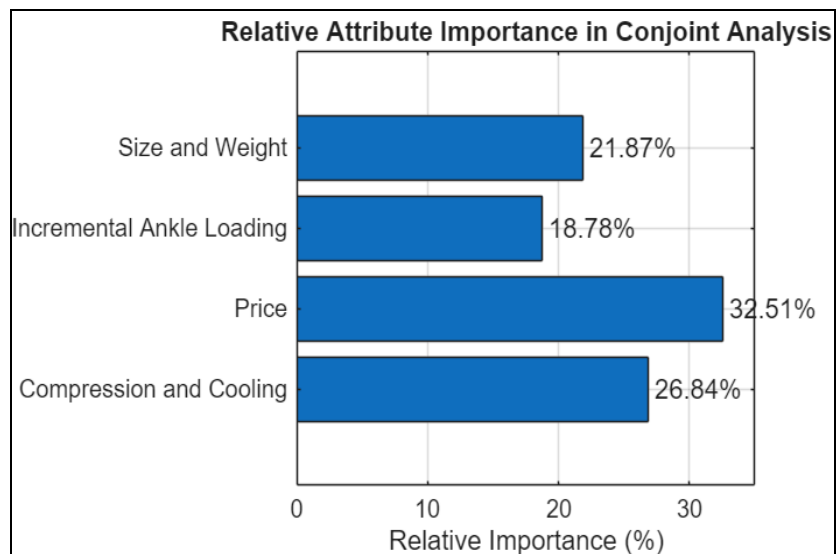


Figure 2: Relative Attribute Importance

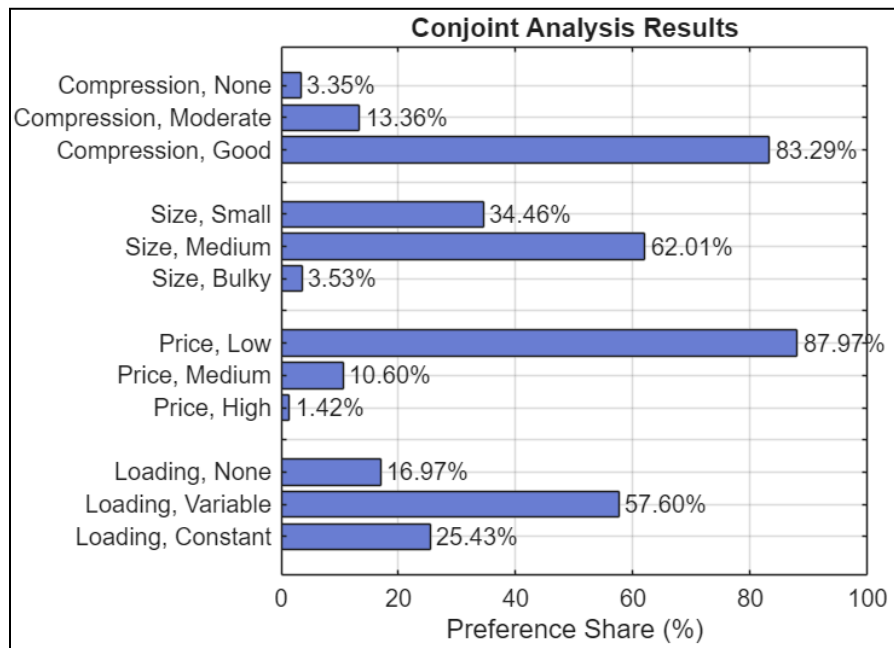


Figure 3: Attributes Preference Comparison

Conjoint Test 2 (Revised)

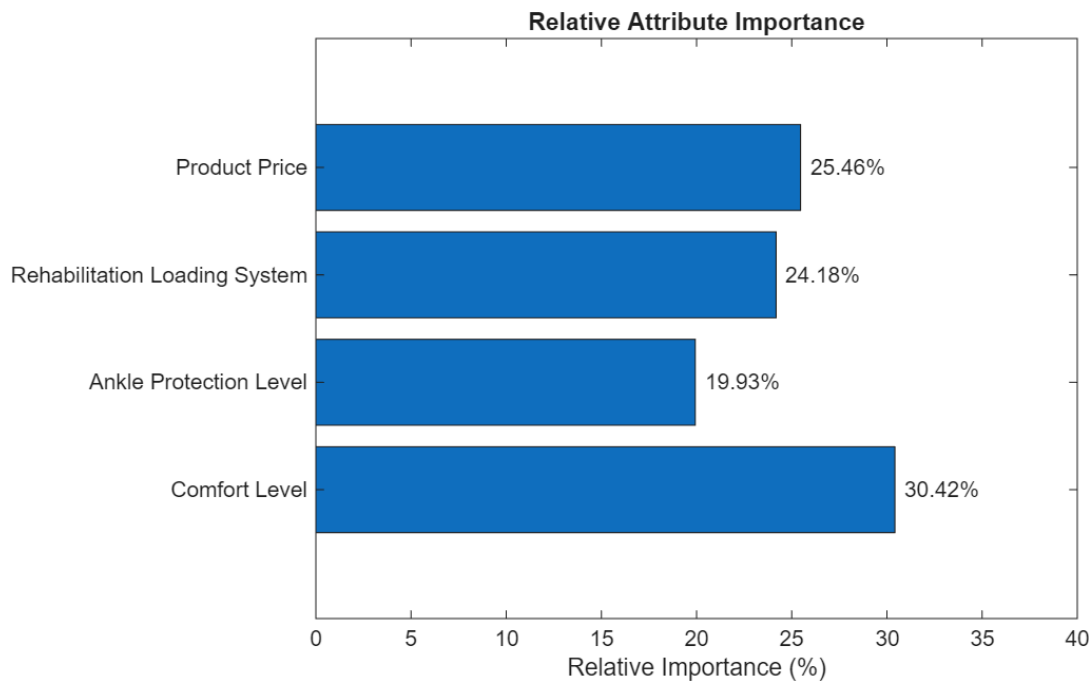


Figure 4: Conjoint v2 Relative Attribute Importance

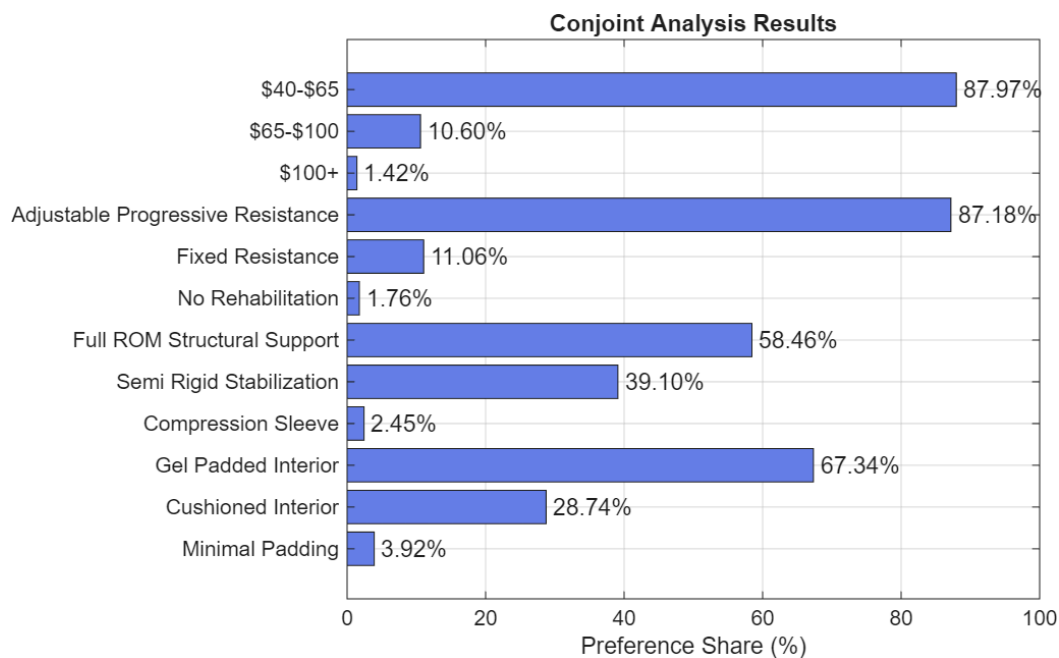


Figure 5: Conjoint v2 Preference Share

Design Impact

To provide a clear understanding of how product features influenced decisions, we first analyzed the results from our initial conjoint analysis.

Figure 2 shows the relative importance of each attribute from the initial conjoint analysis. Price was the most influential factor, accounting for 32.51% of total importance. Compression and cooling accounted for 26.84%, size and weight for 21.87%, and incremental ankle loading for 18.78%. These results indicate that customers are highly cost-conscious, while also valuing comfort-related features such as compression and lightweight design. The lower importance of incremental ankle loading suggests that customers initially prioritize comfort and affordability over advanced rehabilitation functionality.

Figure 3 details preference share percentages for each attribute level from the first conjoint analysis. The low-price option was preferred by 87.97% of respondents, while only 1.42% chose the high-price option, highlighting strong price sensitivity. Good compression was favored by 83.29%, compared to 13.36% for moderate and 3.35% for no compression. Medium-sized designs were preferred at 62.01%, while bulky designs were least favored at 3.53%, indicating a preference for a balance of support, mobility, and comfort. For rehabilitation loading, variable loading was most preferred (57.60%), followed by constant loading (25.43%) and no loading (16.97%). Although rehabilitation functionality was not the top attribute, customers valued the ability to adjust support during recovery.

Following the initial conjoint analysis, customer interviews were conducted to better understand the reasons behind these preferences and to identify additional requirements. Customers consistently emphasized the importance of long-term comfort, ankle protection, recovery speed, and rehabilitation effectiveness. Many wanted a brace that was comfortable for daily use and actively supported progressive recovery and injury stabilization. Insights from these interviews, combined with the initial analysis, directly informed the development of a more refined second conjoint analysis.

Figure 4 shows the relative attribute importance from the second conjoint analysis. Unlike the first analysis, which focused on compression, size, and loading, the second analysis addressed customer requirements identified in interviews: comfort level, rehabilitation loading system, ankle protection level, and price. Comfort level became the most important attribute at 30.42%, followed by price at 25.46%, rehabilitation loading system at 24.18%, and ankle protection at 19.93%. These results indicate a shift in customer priorities. While affordability remained important, customers placed greater emphasis on comfort and rehabilitation performance when considering the brace for real recovery.

Figure 5 presents preference share results from the second conjoint analysis, offering detailed insight into updated customer priorities. The \$40 to \$65 price range remained most preferred at 87.97%, confirming that affordability strongly influences purchasing decisions. Customers also showed a strong preference for adjustable progressive resistance systems, with 87.18% indicating demand for rehabilitation systems that allow gradual increases in ankle loading. For ankle protection, full-range-of-motion structural support was preferred at 58.46%, followed by 39.10% for semi-rigid stabilization and 2.45% for compression sleeves. This indicates customers want effective ankle protection while maintaining mobility. Comfort features were also influential, with gel-padded interiors preferred at 67.34%, compared to 28.74% for cushioned interiors and 3.92% for minimal padding.

The results from Figures 4 and 5 formed the basis for the current LoadStep design. Incorporating interview feedback and refined conjoint analysis, the final product prioritizes ergonomic comfort, progressive rehabilitation loading, structural ankle protection, and recovery-focused support while maintaining affordability.

2.4 Describe Engineering Characteristics (GS)

Table 2: Engineering Characteristics

Engineering Characteristic (EC)	Description	Units
Brace Security	Ability of the brace to remain securely fastened and prevent unwanted movement	Pa
ROM Control	Ability to control and limit ankle range of motion during movement	N·m / °
Compression Level	Amount of pressure applied to the ankle for stability and circulation support	Pa
Cooling Level	Time required to reduce ankle temperature or maintain cooling effectiveness	s
Mass	Total weight of the ankle brace	g
Rehabilitation Loading	The amount of additional resistive torque applied during inversion	N·m

Subsystem Contributions

Table 3: Engineering Features that contribute to each subsystem

<u>Subsystem</u>	Engineering Feature #1	Description: How	Engineering Feature #2	Description: How	Engineering Feature #3	Description: How
Shin / Top Ankle Splint	Brace Security	Keeps ankle stabilized and prevents brace slippage	ROM Control	Limits excessive ankle movement in all directions	Rehabilitation Loading	Transfers controlled load during rehab activity
Over / Under Straps	Brace Security	Secures the brace tightly to the leg and foot	Compression Level	Applies adjustable pressure	ROM Control	Helps restrict unwanted ankle motion through tension
BOA Top Fasteners	Compression Level	Enables precise tightening for consistent pressure	Brace Security	Maintains a secure fit during dynamic movement	ROM Control	Reduces the unintended ankle range of motion

Gel Padding Interior	Compression Level	Distributes pressure evenly to reduce localized stress	Cooling Level	Helps manage heat and improve thermal comfort	Mass	Adds slight weight due to cushioning material
Carbon Fiber “Skeleton”	Mass	Lightweight structure minimizes overall device weight	Brace Security	Provides rigid support to prevent deformation	Rehabilitation Loading	Handles applied forces during movement and rehab
Water-Sol uble Padding	Cooling Level	Provides a cooling effect when activated with moisture	Mass	Minimal structural weight contribution	Compression Level	Conforms to the ankle to improve pressure distribution

Subsystem Interactions Matrix

Table 4: Subsystem Interaction Matrix

<u>Subsystem</u>	Shin/Top Ankle Splint	Over/Under Straps	BOA Top Fasteners	Gel Padding Interior	Carbon Fiber “Skeleton”	Water-Soluble Padding
Shin/Top Ankle Splint	X	+	++	+	-	-
Over/Under Straps	+	X	+	-	--	-
BOA Top Fasteners	++	+	X	+	-	-
Gel Padding Interior	+	-	+	X	--	+
Carbon Fiber “Skeleton”	-	--	-	--	X	-
Water-Solu ble Padding	-	-	-	+	-	X

2.5 Product Architecture (GS)

Functional Decomposition

1. Shin “Splint”

A rigid support that sits along the shin and upper ankle.

- Primary function: Stabilizes the ankle joint and prevents excessive movement
- Role in system: Main structural limiter of motion

2. Gel Cushion

Soft cushioning layer placed between the brace and skin.

- Primary function: Improves comfort and reduces pressure points
- Role in system: Interface between rigid structure and body

3. BOA Tensioning Lines

A cable-based tightening system (dial-controlled).

- Primary function: Evenly tightens the brace around the ankle
- Role in system: Precision adjustment mechanism

4. Overhead Strap + Velcro

External securing strap across the top of the brace.

- Primary function: Locks the brace in place during movement
- Role in system: Secondary retention system for safety and stability

5. Carbon Fiber “Skeleton”

Lightweight semi-flexible structural frame.

- Primary function: Provides high strength with minimal weight
- Role in system: Load-bearing backbone of the brace

6. Cloth

Flexible fabric layer forming the base or inner sleeve.

- Primary function: Comfort interface and structural foundation for attachments
- Role in system: Wraps and holds subsystems together

7. Resistance Bands

Elastic components used for movement assistance or resistance.

- Primary function: Adds controlled resistance or support during motion
- Role in system: Rehabilitation training element

Clustered Module Schematic

A clustered module schematic of the LoadStep with part numbers appears in Figure 6.

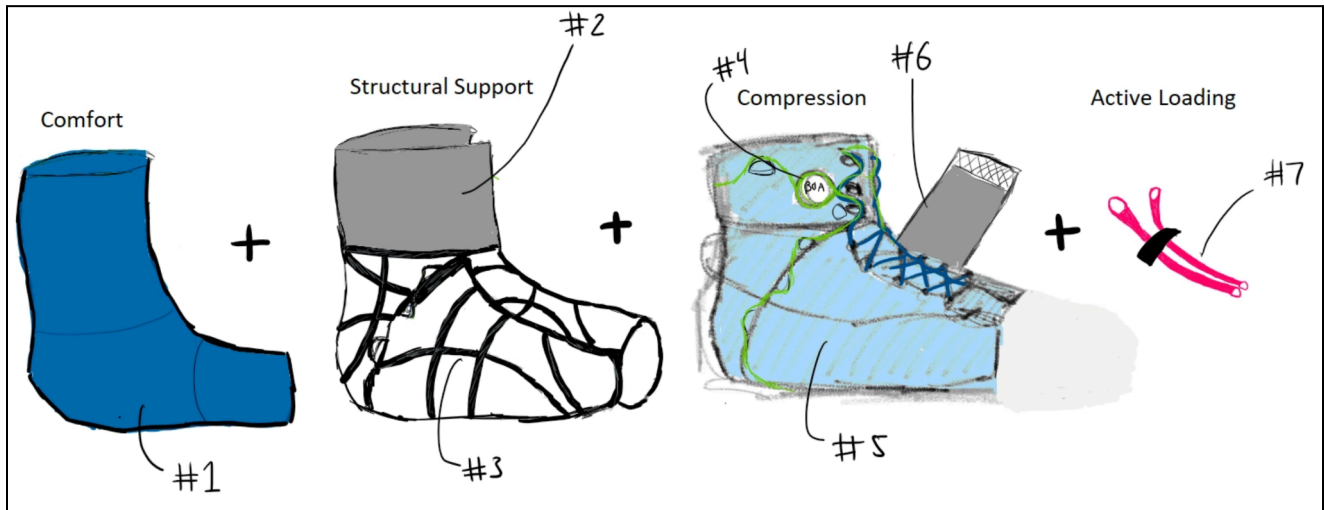


Figure 6: Clustered Module Schematic for LoadStep

Breaking up the product into its different modules:

- (1) Comfort Interface Module → #1 Gel cushion (inside)
- (2) Structural Support Module → #2 Shin splint + #3 Carbon fiber skeleton
- (2) Compression System Module → #4 BOA lines + #5 Cloth (inside), #6 External Straps
- (4) Rehab/Active Control Module → #7 Resistance Bands

Geometric Layout

The final geometric layout of LoadStep is featured in Figure 7.

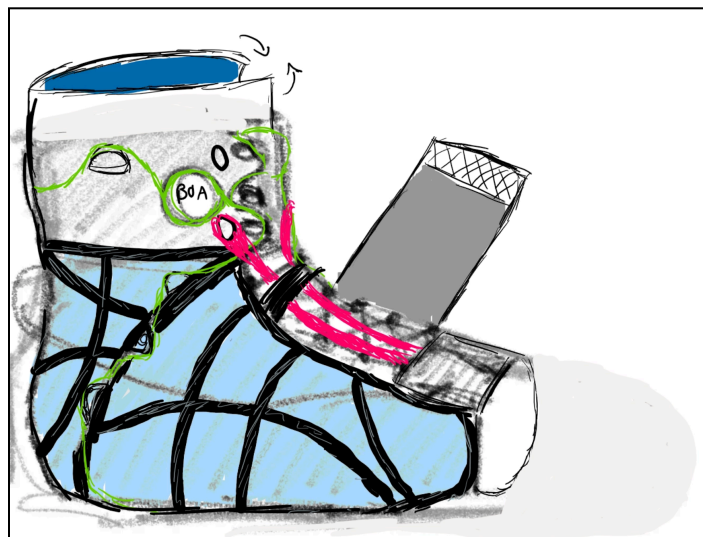


Figure 7: Final Geometric Layout

Module Interfaces

- **Spatial**

- **Gel + Splint interacts**

- The gel layer is shaped to conform closely to the contours of the ankle and shin, filling small anatomical gaps between the rigid splint and the skin. This ensures even contact pressure, reduces localized stress points, and improves comfort during movement by distributing loads across a larger surface area.

- **Outer shell + cloth + gel fit inside the shoe**

- The outer carbon fiber shell, cloth lining, and gel padding are designed as a low-profile layered stack, where each component follows the natural taper of the ankle. The materials are selected and shaped to minimize outward expansion, allowing the brace to sit within standard shoe dimensions while maintaining structural integrity and cushioning without adding excessive volume.

- **How adjustable lines ride along the shape of the foot to provide tension**

- The BOA cables or adjustable tension lines are routed along the natural curvature of the foot and ankle, following anatomical pathways that allow even tightening. This routing ensures that applied tension is distributed circumferentially rather than concentrated at single points, improving stability while avoiding pressure hotspots and maintaining consistent compression during motion.

- **Energy**

- **How the tension in the lines is stored and what motions they resist**

- The BOA tensioning lines store mechanical energy as elastic tension is applied during tightening. When the dial system pulls the cables, the energy is distributed along the routing path and held in static tension, allowing the system to maintain consistent compression without continuous input force.
 - This stored tension resists motion by creating opposing forces against ankle movement, particularly limiting inversion, eversion, and excessive plantarflexion/dorsiflexion. As the foot moves, the tension lines counteract displacement by converting that motion into increased cable strain, effectively providing passive resistance and stabilizing the joint throughout dynamic activity. Therefore, loosening the BOA system allows a greater range of motion.

- **How to adjust torque:**

- Adjustment is achieved by changing the attachment height and routing position of the resistance bands along the brace. This directly alters the direction of the applied force vector relative to the ankle joint.
 - Counter-torque is generated primarily by the vertical (y-direction) component of the band tension (Figure 8). A force in the x-direction creates no moment. By attaching the resistance bands higher on the structure, a greater portion of the tension force is aligned with the y-axis, increasing the effective moment arm contribution. As a result, a larger torque is produced about the ankle joint, requiring more effort to overcome, thereby increasing the system's resistance.

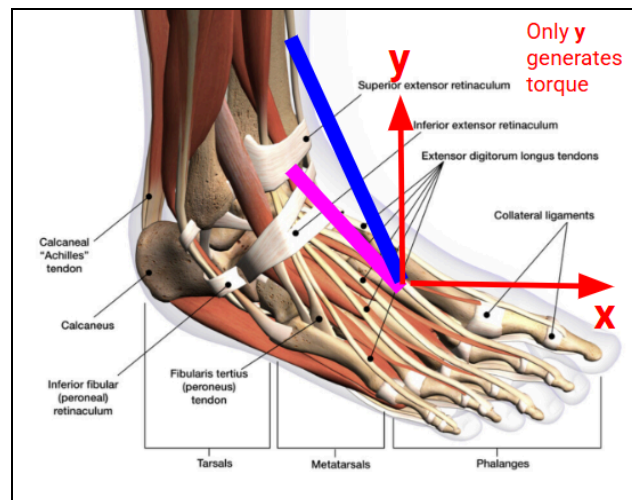


Figure 8: Torque generation diagram

- **Information**

- Information flow occurs through user-adjustable settings (BOA rotational orientation, strap position, resistance band anchor location) and sensory feedback from discomfort, stability, and mobility.

- **Material**

- **Protection/flexibility balance:**

- The ankle brace system uses a combination of rigid composite materials and compliant cushioning to balance support and comfort (Figure 9). The carbon fiber “skeleton” provides high stiffness and strength at low mass, ensuring structural stability while minimizing weight and preventing unwanted deformation under load.

- **Thermal Comfort and Moisture Isolation**

- The gel layer is positioned under the foot to serve as both a cushioning interface and a thermal barrier between the skin and the outer cloth layer. This separation prevents moisture from the external cloth layer from directly contacting the skin while still allowing heat transfer through the gel material (Figure 9).
- The gel is designed to maintain thermal conductivity for cooling performance while limiting moisture permeability, ensuring that cooling can occur without creating a wet or uncomfortable surface against the foot (Figure 9). For optimal performance, the gel should balance high thermal conductivity with low fluid transfer, enabling effective heat dissipation without compromising dryness or comfort.

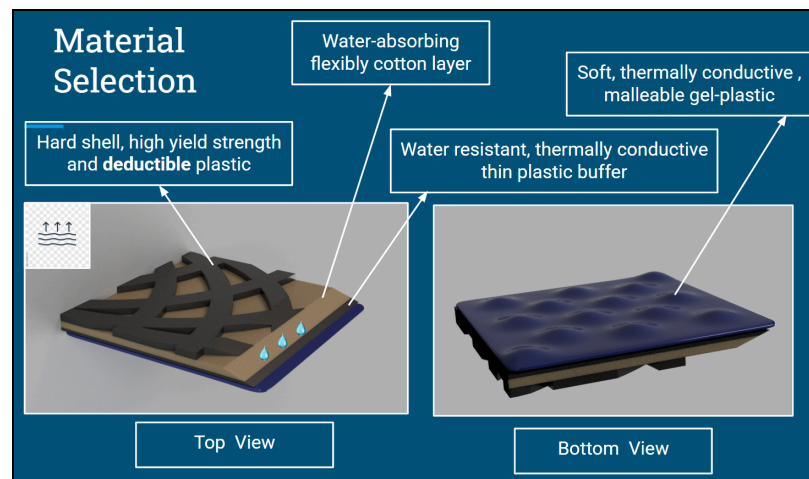


Figure 9: LoadStep Material Make-up

Architecture Type

The **hybrid architecture** combines this modularity with an **integral** structural core, where the rigid frame provides stability, ROM control, and load support. Together, this balance ensures both high mechanical performance and adaptability across different users and conditions.

1. Modular Components

- **BOA tensioning system** - adjustable compression and fit control
- **Over/under straps (Velcro)** - interchangeable fastening and sizing adjustment
- **Gel cushioning layer** - replaceable for comfort and hygiene
- **Resistance bands** - customizable stiffness for different active loading levels

Justification:

These components are designed for personalization, maintenance, and variability in rehabilitation requirements. Modularity allows the brace to adapt to different users and injury stages without requiring a full system redesign.

2. Integral Components

The structural core of the brace is designed as an integrated system where components function as a single load-bearing unit.

- **Carbon fiber “skeleton”**
- **Shin/splint structure (attached frame geometry)**

Justification:

These elements are optimized together to control the range of motion and distribute loads efficiently. Because they directly define structural rigidity and force transfer, separating them would reduce performance and increase the risk of re-injury.

2.6 House of Quality Level 2 (EW)

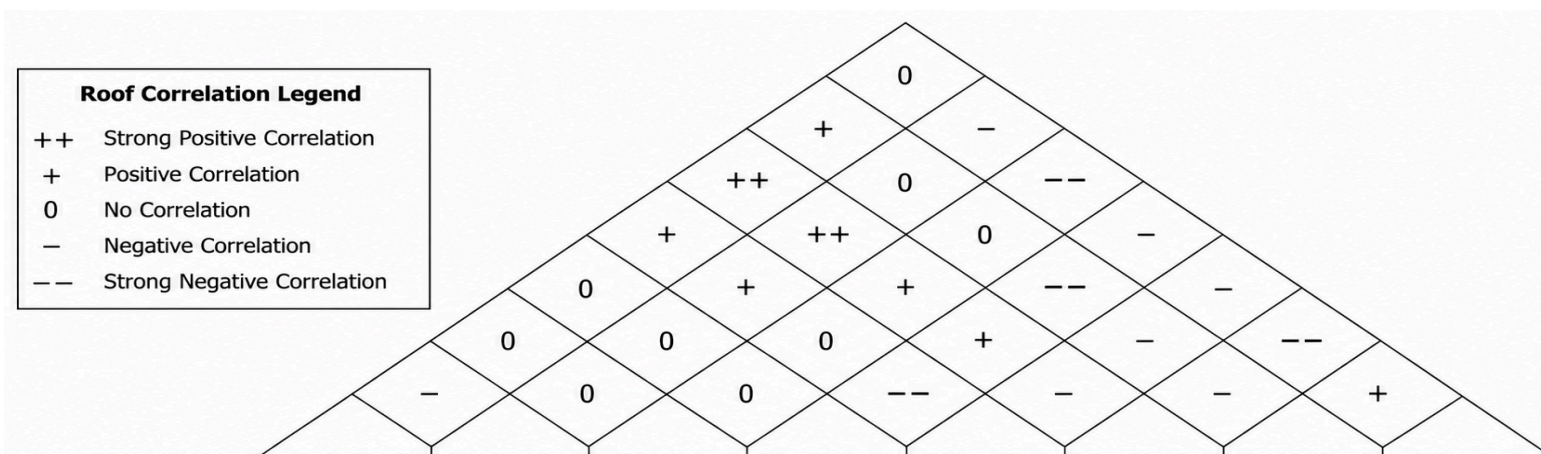
House of Quality Level #2 was generated to show how the engineering characteristics (EC's) map to the part characteristics (PC's) (Table 5). Each part requirement has a corresponding target value. PC1, the shin support stiffness, must be able to resist 150N·m of inversion because that is the maximum amount of torque the ankle experiences during inversion.² PC2, the BOA tension, must be between 40-80 N to ensure secure brace fixation while avoiding excessive compression discomfort. PC3, the resistance fiber loading stiffness, should have a k value between 4,000 and 1,000 Nm (see calculations in Section 2.11). PC4, The outer shell thickness should be between 2 and 4 mm to balance structural rigidity, durability, manufacturability, and overall mass. PC5, the compressive interface pressure, should be between 20 and 30 mmHg to offer relief without constricting blood flow.³ PC6, the water retention for cooling, needs to be 18g to satisfy the 2-hour target cooling timeframe. This follows the formula:

$$E = h_m \cdot A \cdot (C_s - C_{inf})$$

where $h_m = 0.005$ m/s (mass transfer coefficient), $A = 0.05$ m² (exposed surface area), $C_s = 0.02$ kg/m³ (water vapor at wet surface), and $C_{inf} = 0.01$ kg/m³ (ambient water vapor).⁴ This yields an $E = 0.15$ g/min evaporation rate. To sustain evaporative cooling for roughly two hours during rehabilitation activity, 18g of water is required. PC7, the ventilation hole area fraction, must be about 20% to ensure the brace is breathable and will still have structural integrity. Lastly, PC8, the thermal conductivity of the gel, must be greater than 0.3 W/m·K to ensure the gel can distribute and retain the cooling effect inside the brace.

Similar to HOQ Level #1, HOQ Level #2 assigns 9 when the part characteristic highly corresponds to the engineering characteristic, 3 when there is some relationship, 1 when there is little relationship, and 0 when there is no relationship. To calculate the raw score, we multiplied the matrix values by the engineering characteristic weight from HOQ Level #1 and summed the results for each product characteristic. We then converted the values into relative weights and ranked them.

Table 5: House of Quality Level #2



	Part Characteristics	PC1: Shin Support Stiffness [Pa]	PC2: BOA Tension [N]	PC3: Resistance Fiber Loading Stiffness [N/m]	PC4: Outer Shell Thickness [mm]	PC5: Compression Interface Pressure [Pa]	PC6: Water Retention for Cooling [g]	PC7: Ventilation Hole Area Fraction [% open area]	PC8: Thermal Conductivity of Gel [W/m·K]
	Target Value	Resist 150 N·m of Inversion	40-80 N	4,000 - 11,000 N/m	2-4 mm	20-30 mmHg	≥ 18g	~20%	≥ 0.3 W/m·K
Engineering Characteristics	Weight: HOQ Level #1	Matrix Values							
EC1: Brace Security [Pa]	21.06%	9	3	3	3	3	0	1	3
EC2: ROM Control [N·m / °]	17.02%	9	9	9	3	1	0	0	0
EC3: Compression Level [Pa]	16.38%	0	9	1	3	9	0	1	3
EC4: Cooling Level [°C]	15.96%	0	0	0	0	0	9	9	9
EC5: Mass [g]	14.89%	3	1	1	9	1	9	3	3
EC6: Rehabilitation Loading [N·m]	19.5%	0	0	9	1	3	0	0	0
Absolute Weight		3.874	4.068	4.231	3.169	3.010	2.777	2.258	3.006
Relative Weight		14.84%	14.49%	16.21%	12.14%	11.53%	10.64%	8.65%	11.51%
Rank Order		3	2	1	4	5	7	8	6

From this table, it is evident that resistance fiber loading stiffness (PC3), BOA tension (PC2), and shin support stiffness (PC1) are our most important characteristics to make sure to integrate into our design. This will ensure the brace functions as intended and meets our engineering and customer requirements.

It is important to note that the highest-ranking part characteristics from HOQ Level #2 do not necessarily align with the highest-ranking engineering characteristics from HOQ Level #2 (e.g., loading

fiber resistance was a top-ranked part characteristic but not an engineering characteristic). This occurs because part characteristics represent specific physical components of the design, while engineering characteristics represent broader system-level performance metrics. Therefore, an individual subsystem, such as the resistance-loading fibers, may be highly important on its own, even if its effects are distributed across multiple engineering characteristics (e.g., ROM control, ankle stability, and rehabilitation loading).

2.7 Requirements Traceability Matrix (RTM) (EW)

To map customer requirements (CR's) to engineering (EC's) and part (PC's) characteristics, a requirements traceability matrix (RTM) was developed (Table 6). The table describes how each mechanism in the device successfully satisfies its design constraints.

Table 6: Requirements Traceability Matrix

Customer Requirement	Engineering Characteristic	Part Characteristic	Mechanism
CR1 Comfortable	EC3 Compression Level; EC4 Cooling Level; EC1 Brace Security; EC5 Total Mass	PC1 Shin Support Stiffness; PC5 Compression Interface Pressure; PC6 Water Retention for Cooling; PC8 Thermal Conductivity of Gel; PC2 BOA Tension	Soft, cushioning materials to distribute pressure; User tightens laces to toggle compression; Cold therapy from the wet layer, as the water evaporates, keeps the user comfortable throughout healing
CR2 Cost Friendly	EC5 Mass; EC1 Brace Security	PC4 Outer Shell Thickness; PC8 Thermal Conductivity of Gel; PC1 Shin Support Stiffness	Cheap enough to manufacture and assemble the materials despite multiple layers; high stiffness and specific thermal conductivity parts could drive up cost, in particular
CR3 Ankle Protection	EC1 Brace Security, EC2 ROM Control; EC6 Rehabilitation Loading,	PC3 Resistance Fiber Loading Stiffness; PC1 Shin Support Stiffness; PC2 BOA Tension	User decides how much ROM to keep versus restrict via the BOA system; resistance fibers accelerate healing, ultimately protecting the ankle from further injury
CR4 Cold Therapy	EC4 Cooling Level	PC6 Water Retention for Cooling; PC8 Thermal Conductivity of Gel	Ventilation evaporates wet midsection to cool ankle; thermally conductive plastic interfaces with water-absorbing cotton layer to keep ankle cool; overall helps distract user from feeling pain
CR5	EC2 ROM	PC2 BOA Tension; PC3	Velcro, Laces, and BOA are somewhat

Installation Time	Control, EC6 Rehabilitation Loading	Resistance Fiber Loading Stiffness	quick to fasten systems, decreasing installation time; higher installation time than competitors
CR6 Light Weight	EC5 Mass	PC4 Outer Shell Thickness; PC6 Water Retention for Cooling; PC7 Ventilation Hole Area Fraction	Thin hard plastic shell, thermally conductive plastic, and water-absorbing cotton layer are all lightweight materials; perforated holes in the material decrease weight, too
CR7 Breathable	EC4 Cooling Level	PC7 Ventilation Hole Area Fraction; PC6 Water Retention for Cooling	The cotton water-absorbing layer is very breathable; the hard shell features holes to increase ventilation
CR8 Recovery Speed	EC6 Rehabilitation Loading	PC3 Resistance Fiber Loading Stiffness; PC1 Shin Support Stiffness	Progressive loading supports healing as tissue adapts to mechanical stress

A table was also developed to identify design gaps in the system, or where our device lacks part characteristics to fully meet the engineering characteristics and customer requirements (Table 7).

Table 7: Design Gaps

Design Gaps	Explanation
Installation Time	Installation time is currently indirectly represented by BOA tension and resistance loading stiffness, but there is no dedicated PC that specifically quantifies fastening efficiency.
Quantitative Cooling Characterization	Cooling performance is represented through water retention and thermal conductivity, but no PC currently captures cooling duration or temperature reduction over time.
Durability/Fatigue Behavior	Long-term wear and cyclic loading performance of resistance fibers and shell materials are not directly represented in the current PCs.
Fit Adaptability Representation	Current PCs do not directly quantify how the brace accommodates different ankle geometries or shoe sizes other than by tightening various straps.
Direct Airflow Metric	Ventilation hole area fraction approximates breathability, but airflow itself is not directly measured.

Similarly, a table was developed to explore the scope creeps in the system, or where part characteristics feel a bit broad or likely to be misinterpreted (Table 8).

Table 8: Scope Creeps

Scope Creeps	Explanations
PC8 Thermal Conductivity of Gel	Strongly supports cooling and comfort, but has weak linkage to other customer requirements, making it specialized relative to the broader stabilization/healing-focused system.
PC7 Ventilation Hole Area Fraction	Supports breathability and weight reduction, but excessive focus on perforation optimization may contradict the device's primary rehabilitation and stabilization objectives.
PC1 Shin Support Stiffness [Pa]	Supports force redirection to the shin to avoid excessive load on the ankle while still providing some degree of loading to accelerate healing. However, the stiffness of this shin support is only relevant in providing a strong anchor for these resistance fibers. It enables load transfer rather than directly immobilizing the ankle joint itself.

2.8 TRIZ Analysis (GS)

Engineering Contradictions

- **Toughness vs Comfort:** A stiffer material is more stable but less comfortable. We are using a ductile material as a midpoint between the two.
- **Cost vs Features:** We sacrificed the cost-friendly design requirement in favor of features that improve user comfort (cooling materials and compression) and promote healing (resistance fibers).
- **Breathability vs Stability:** We sacrificed breathability to include a hard plastic layer for stability. The carbon fiber skeleton still allows the device to remain a bit breathable.
- **Cold Therapy vs Lightweight:** When wet, the cotton layer will add weight to the device; however, evaporative cooling for user comfort is a higher priority.
- **Installation Time vs Adjustability:** Laces, velcro, and the BOA tensioning system take time to adjust, but adjustability is very important to vary ROM and degree of compression based on user needs. The BOA system is faster than velcro alternatives.

2.8.1.2 Substance Field Analysis

Table 9 showcases the substance field analysis for LoadStep.

Table 9: Substance Field Analysis

#	Problem / Function	Contradiction (What improves vs what gets worse)	Substance 1 (S1)	Substance 2 (S2)	Field (F)	Proposed Solution
1	Increase range of motion control	Restricting motion can make walking uncomfortable	Splint frame	Ankle joint	Mechanical constraint force	Restrict only dangerous motion directions while allowing natural gait
2	Increase ankle stability during movement	More stability increases mass and reduces comfort	Carbon fiber splint	Ankle joint	Mechanical force (support/torque)	Use a carbon fiber structure for high stiffness with minimal mass
3	Improve compression for support	Higher compression reduces comfort and circulation	BOA cable system	Leg/foot	Tensile force	Use dial-based BOA system for precise, adjustable compression
4	Improve comfort and pressure distribution	More cushioning increases bulk and heat retention	Gel padding	Skin /sock	Pressure distribution / thermal contact	Use a thin viscoelastic gel to spread the load without excessive thickness
5	Reduce overall brace weight	Lower weight can reduce durability and protection	Carbon fiber skeleton	Entire brace system	Structural load transfer	Use hollow carbon fiber geometry to maintain strength while reducing mass

Figure 10 showcases an additional graphic for visualization for substance field analysis example #1 (Table 9):

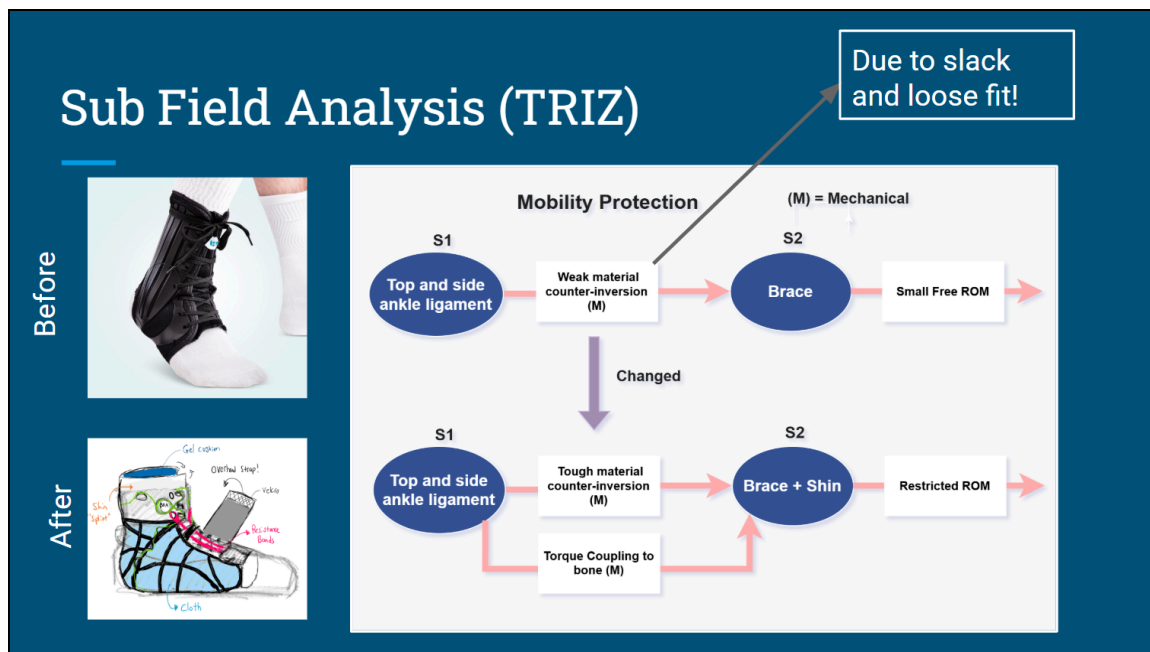


Figure 10: Substance Field Analysis graphic for example #1

Historical TRIZ Solutions

1. **Principle #17, Moving to a new dimension:** We moved the system beyond its boundary by connecting to the shin for inversion restriction, creating a larger moment arm for better torque restriction rather than increasing the system's rigidity.
2. **Principle #31, Use of a porous material:** We used a carbon fiber skeleton with gaps between the pieces to improve breathability while still providing a stabilizing, rigid component.
3. **Principle 15, Dynamicity:** We allowed the system to conform to optimal conditions for each user. The user can change the anchor point for the resistance loading fibers to toggle the degree of loading applied to their ligaments. They can also adjust the BOA system, laces, and velcro to toggle the degree of compression and ROM restriction.

Proposed Solution

Given the complexity of this product and the many interconnected pieces that need to be optimized, the best way to proceed with the proposed solution is to test using rapid prototyping. It is unclear how large/heavy an ankle brace can be before it becomes unpreferred to a customer, just as with cooling intensity, duration, impact resistance, and overall stiffness. All components will remain present; however, we will need to complete extensive customer-facing validation testing to finalize the exact specifications for each component.

Embodiment

Figure 11 showcases a flow diagram representing how to use and implement the product throughout the stages of ankle progression during rehabilitation:

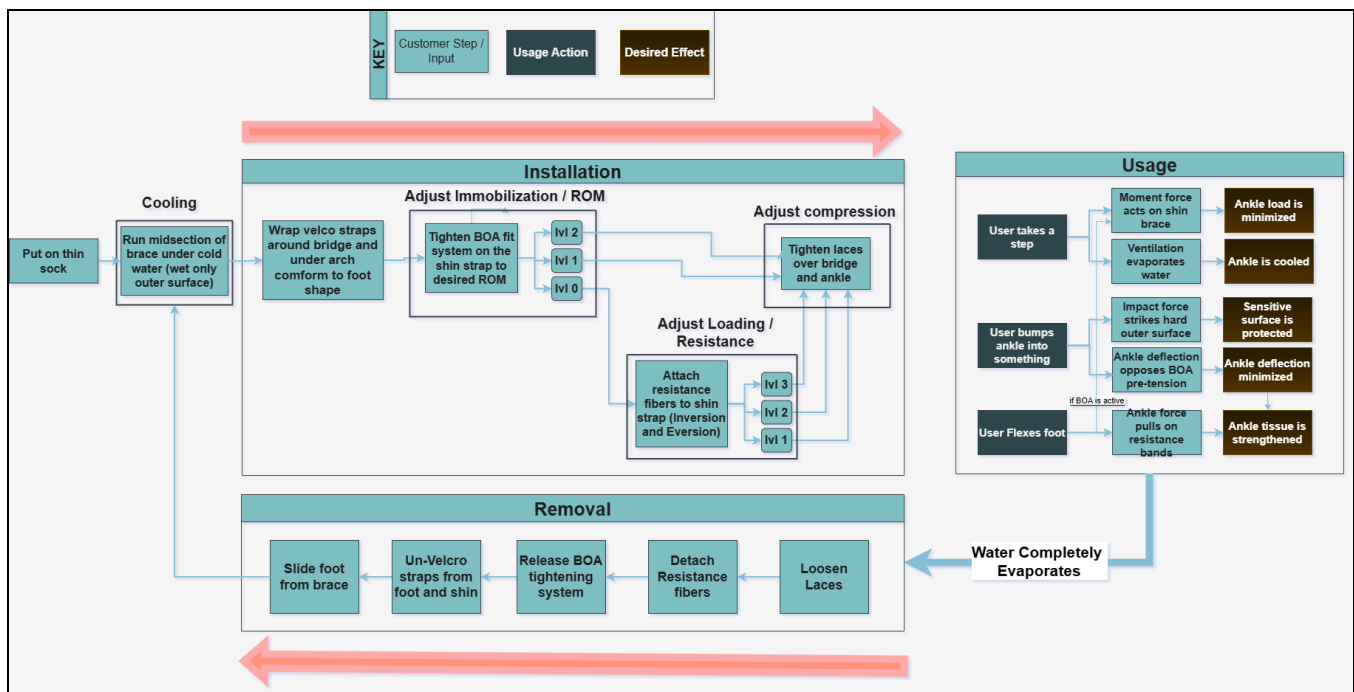


Figure 11: LoadStep flow diagram

2.9 What is the embodiment of your solution? (GS)

As listed in HOQ Level #2 (Section 2.6, Table #5), our part requirements are as follows:

1. **Shin Support Stiffness** [Pa]
2. **BOA Tension** [N]
3. **Resistance Fiber Loading Stiffness** [N/m]
4. **Outer Shell Thickness** [mm]
5. **Compression Interface Pressure** [Pa]
6. **Water Retention for Cooling** [g]
7. **Ventilation Hole Area Fraction** [% open area]
8. **Thermal Conductivity of Gel** [W/m·K]

After discussing among ourselves and consulting with Prof. Shepard, our team decided that not much additional information would be gained if we pursued a CAD or physical model of our product. The product is too complex and contains too many parts with patient-specific, anatomy-dependent geometry to model easily in CAD. Refer to the sketch to understand the “Works Like” model in Figure 1 and the embodiment of our solution.

2.10 Traceability Matrix (EW)

A traceability matrix was created to further detail how each customer requirement maps to each subsystem and its underlying mechanisms (Table 10).

Table 10: Traceability Matrix

Customer Requirement	Subsystem	Mechanism
CR1: Comfortable	Material choices; Adjust compression	Soft, cushioning materials to distribute pressure; User tightens laces to toggle compression; Cold therapy from the wet layer, as the water evaporates, keeps the user comfortable throughout healing
CR2: Cost Friendly	Material Selection	Cheap enough to manufacture and assemble the materials despite multiple layers; high stiffness and specific thermal conductivity parts could drive up cost, in particular
CR3: Ankle Protection	Adjust Immobilization/ ROM	User decides how much ROM to keep versus restrict via the BOA system; resistance fibers accelerate healing, ultimately protecting the ankle from further injury
CR4: Cold Therapy	Cooling; Material Choices	Ventilation evaporates wet midsection to cool ankle; thermally conductive plastic interfaces with water-absorbing cotton layer to keep ankle cool; overall helps distract user from feeling pain
CR5: Installation Time	Fastening Decisions	Velcro, Laces, and BOA are somewhat quick to fasten systems, decreasing installation time; higher installation time than competitors
CR6: Light Weight	Material Choices	Thin hard plastic shell, thermally conductive plastic, and water-absorbing cotton layer are all lightweight materials; perforated holes in the material decrease weight, too
CR7: Breathable	Material Choices	The cotton water-absorbing layer is very breathable; the hard shell features holes to increase ventilation
CR8: Recovery Speed	Adjust Loading / Resistance	Progressive loading supports healing as tissue adapts to mechanical stress

2.11 Mechanical Analysis (GS)

Analytical Calculations

Literature demonstrates that substantial ankle joint moments occur during gait and athletic activity.² Based on these loading conditions and the intended use during dynamic athletic motion, the LoadStep was designed to resist inversion torques on the order of 150 N·m. We propose doing so by using high-strength materials in compression, including a shin splint and a carbon fiber outer skeleton.

Finite Element Analysis (FEA)

- **Static loading** (hand calculations completed due to model complexity)

Assumptions made in this model:

- Resistance bands have a constant spring constant with displacement
- Reaction force can be modeled as an ideal spring
- The pin joint of the ankle is infinitely small
- Angular displacement is constant: 20°

Target torque generated by the foot during a 100kg person's stride: 200 Nm

Figures 12 and 13 illustrate variables " θ ," " k ," and " x " in a free-body diagram. " θ " is the angle between a horizontal line drawn at the ankle and the position of the toe. " k " is the stiffness of the resistance loading fibers. " x " is the displacement when the foot is flexed.

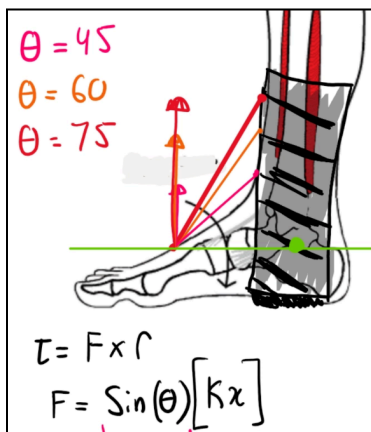


Figure 12: Free Body Diagram #1

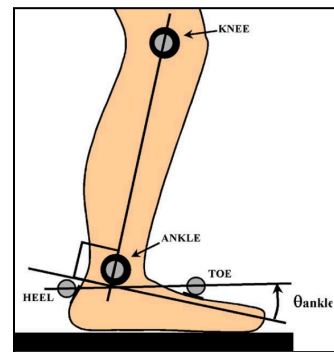


Figure 13: Free Body Diagram #2

Moment Arm (distance from center of pressure to where the bands are tied): $r = 0.18\text{m}$

Displacement (when foot is flexed): $x = 0.05\text{m}$

Using the formulas in Figure 12:

- $k_{\theta=30^\circ} = 4,000 \text{ N/m}$
- $k_{\theta=45^\circ} = 6,250 \text{ N/m}$
- $k_{\theta=60^\circ} = \sim 11,000 \text{ N/m}$

These values are much stiffer than traditional bands. Materials with k values of this order of magnitude are latex rubber and thermoplastic elastomers (TPE plastics).

• **Cyclic loading**

Since latex rubber and thermoplastic elastomers are commonly used in cyclic loading applications due to their high fatigue resistance, the resistance-loaded fibers are expected to withstand

prolonged use at relatively low strain levels.⁶ Failure would only occur if the bands are stretched far beyond their strain limits, which would not be possible within the framework of LoadStep.

2.12 Project Timeline (GS)

Table 11 displays the Gantt Chart for this semester.

Table 11: Semester Gantt Chart

Assignment / Milestone	Feb	Mar	Apr	May
Problem Identification & Research	X			
Presentation 1 – Hypothesis	X			
Initial Concept Development	X	X		
Presentation 2 – PDR		X		
Product Architecture & Embodiment Design		X	X	
Presentation 3 – Embodiment & Product Architecture			X	
PDS V1 (<i>Due Apr 23</i>)			X	
Prototype Refinement & Testing			X	X
Final PDS (<i>Due May 16</i>)				X
Final Patent (<i>Due May 16</i>)				X
Final Presentation Preparation				X
Final Presentation				X

3. Market Identification

3.1 Target Market (CL)

LoadStep’s primary target market is student athletes, with an initial focus on Cornell, including club and recreational participants. These athletes often engage in sports that involve cutting, jumping, and rapid directional changes, which heighten the risk of ankle injuries. More broadly, the market includes millions of high school and collegiate athletes nationwide. The core demographic is individuals aged 14 to 24 who are physically active, performance-driven, and motivated to recover quickly. Their

need to balance academics and athletics increases demand for recovery solutions that fit into daily routines.⁷

The market is both large and expanding. In the United States, over 8 million high school athletes and more than 500,000 NCAA athletes represent a significant population at risk for sports-related injuries. Between 2015 and 2019, high school athletes experienced approximately 5.2 million injuries⁷, while NCAA data reported about 1.05 million injuries across 25 collegiate sports over five years, averaging more than 210,000 injuries annually.⁸ Ankle injuries account for 16 to 40 percent of all sports-related injuries, and nearly 85% of these are classified as acute lateral ankle sprains.³

To estimate the Serviceable Addressable Market (SAM), a midpoint of 30% was used for ankle injuries. This results in approximately 1.63 million ankle-related injuries. With 85% of these classified as acute lateral ankle sprains, the estimated Serviceable Addressable Market for LoadStep is about 1.39 million cases among high school and collegiate athletes. Additionally, the global sports and outdoor equipment market exceeds \$226 billion, with strong demand for performance, recovery, and rehabilitation products, indicating significant long-term growth potential for LoadStep.

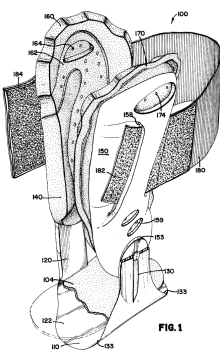
3.2 Consumer Behavior (CL)

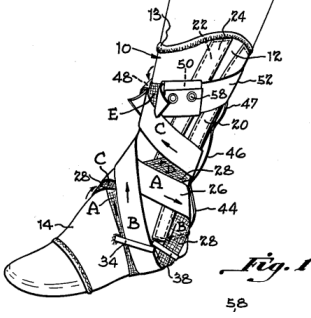
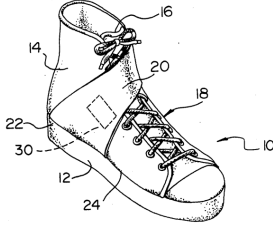
Student athletes who experience ankle sprains frequently encounter challenges during the recovery phase, particularly after mild injuries sustained in games or practice. These injuries are prevalent, accounting for an estimated 16 to 40 percent of all sports-related injuries.⁴ During recovery, users face both physical and performance-related challenges. Pain and swelling can restrict walking and daily activities, while reduced mobility may decrease participation in team activities and result in lost playing time. Consequently, athletes prioritize solutions that reduce discomfort, support mobility, and accelerate recovery. Traditionally, athletes have relied on methods such as taping, standard ankle braces, ice, and physical therapy. Although these solutions provide some support, they often lack convenience, comfort, or adaptability. Taping requires frequent reapplication, traditional braces can be bulky, and ice therapy is not easily incorporated into daily movement. These limitations highlight the need for a single product that combines support, comfort, and functional recovery.

3.3 HOQ: Again (EW)

Table 12 outlines our primary competitors for LoadStep and provides a brief description of each product. A common theme among competitors is that they are geared towards providing the user with comfort as their ankle heals, through cooling and/or compression. No competitor products introduce loading to simultaneously promote healing.

Table 12: Competitors for LoadStep

Competitor Product	Image	Description
A) Med Spec ASO Speed Lacer		Product A is a standard ankle brace that provides compression through laces and straps. The goal of this product is to provide ankle support for sports injury recovery.
B) Ankle Support Brace		Product B is a lightweight brace with highly adjustable straps. The product applies compression, but not too much rigid support, to the ankle. The goal of this product is to provide relief for ankle swelling.
C) Comfpack Ankle Ice Pack Wrap		Product C is an ankle wrap with an integrated ice pack. This product focuses less on providing compression or support for the ankle and instead uses cooling to mitigate pain. The product is placed in the freezer to get cold, and it lasts for 2 hours.
D) Active Ankle T2 Ankle Brace		Product D prevents inversion and eversion (rolling of the ankle) with rigid shin supports. Simultaneously, it promotes agility through a hinge. The device contains padding to support the ankle throughout long-term wear. It has a discrete enough profile to fit under a sneaker.
E) Ankle Support Brace (US5971946A) ⁹		Product E prevents inversion and eversion by using tall, rigid supports that extend from a foot plate along both sides of the ankle. Hinged medial and lateral side panels allow controlled ankle motion while maintaining stability. The brace includes ventilated, padded side supports to improve comfort and airflow during long-term wear. Cleats on the bottom of the foot plate help stabilize the foot during athletic activity.

F) Biomechanical Ankle Brace (US4844058A) ¹⁰		Product F uses an elastic sleeve with adjustable elastic straps that wrap around the ankle to provide targeted support. The straps are aligned with specific ligaments and muscles to assist ankle stabilization and rehabilitation. The flexible design allows for comfortable movement while supporting injured soft-tissue structures.
G) Ankle Brace (US3073305A) ¹¹		Product G uses an elastic ankle sleeve with wrapping support straps that stabilize the ankle and restrict excessive heel rolling, which can lead to sprains. The adjustable strap configurations mimic athletic taping techniques while avoiding common tape issues, including slipping, skin irritation, and difficult removal. The flexible brace design supports injured ankles while still allowing athletic movement and repeated use.
H) Athletic Shoe with Ankle Support Strap (US4441265A) ¹²		Product H integrates ankle support directly into a high-top athletic shoe using a tensioned support strap that wraps across the foot and around the ankle. The adjustable strap helps resist inversion while still allowing normal foot biomechanics. Velcro fasteners allow customizable support and quick adjustment during activity.

HOQ1 is necessary to compare our product with competitors and evaluate whether it better meets customer requirements. For reference, here is HOQ 1 again, but it is further described in section 2.1 (Table 13):

Table 13: House of Quality Matrix Level #1 (repeated for clarity)

Engineering Characteristics							
Improvement Direction		↑	↑	↑	↑	↓	↑
Units		Pa	N·m / °	Pa	s	g	N·m
Customer Requirements	Weight Factor	EC1: Brace Security	EC2: ROM Control	EC3: Compression Level	EC4: Cooling Level	EC5: Mass	EC6: Rehabilitation Loading
CR1: Comfortable	3	9	3	9	9	0	3
CR2: Cost-friendly	2	0	0	3	3	9	1

CR3: Ankle protection	4	9	3	3	0	3	3
CR4: Cold therapy	2	0	0	3	9	0	0
CR5: Installation Time	2	3	1	3	3	3	1
CR6: Light weight	4	3	3	1	3	9	0
CR7: Breatheable	1	3	0	3	3	3	0
CR8: Recovery Speed	5	3	9	1	1	0	9
Raw Score		99	80	69	77	75	70
Relative Weight		21.06%	17.02%	14.68%	16.38%	15.96%	14.89%
Rank Order		1	2	6	3	4	5

Using the same customer requirements and relative weights from the HOQ 1 analysis, we calculated our product's performance against competitors (Table 14). A score of 5 indicates excellent performance, 3 indicates okay performance, and 1 indicates poor performance. These scores were multiplied by the weighting factors for the customer requirements and summed to create a score for each product.

Table 14: Competitor Analysis versus LoadStep

Customer Requirements	Weight	Rankings (1 = Poor, 3 = OK, 5 = Excellent)								
		Load-Step	A	B	C	D	E	F	G	H
CR1: Comfortable	3	4	3	5	2	1	2	5	3	5
CR2: Cost-friendly	2	2	2	4	3	1	1	4	3	4
CR3: Ankle Protection	4	4	3	2	1	3	5	2	4	2
CR4: Cold therapy	2	3	1	1	5	1	1	1	1	1

CR5: Installation Time	2	1	3	3	5	2	4	2	2	5
CR6: Light weight	4	4	4	5	3	5	1	5	3	3
CR7: Breathable	1	2	3	2	4	5	2	2	3	3
CR8: Recovery Speed	5	5	4	2	2	3	1	2	1	1
Raw Scores		79	59	62	51	45	45	54	52	53
Relative Weight		15.80%	11.80%	12.40%	10.20%	9.00%	9.00%	10.80%	10.40%	10.60%
Rank Order		1	3	2	7	8	8	4	6	5

From Table 14, LoadStep best targets our custom needs. This is because LoadStep is the only product that not only mitigates pain from healing but also uses active loading to improve recovery speed. Other products may limit ROM to do this; however, ROM alone is not enough to promote healing, and it is more effective in combination with active loading, which is exactly what LoadStep targets. All the other products vary by only a couple of percentage points, primarily because they all aim to achieve similar goals: good compression and restricted ROM. None of the products on the market target active loading or cold therapy, except for Product C, which is the only one that really accomplishes cold therapy. Products A, B, and F are relatively higher because of their lightweight design, while still providing good compression. Product D is low due to its lack of comfort and slow installation time. Product E is also low since it is a standard boot, which is relatively bulky and limits ROM for ankle healing.

To best calculate a BOM for the product, HOQ Level #2 was constructed, which outlines target values for product requirements and their relative importance to engineering requirements. For reference, here is HOQ 2 again, but it is further described in section 2.6 (Table 15):

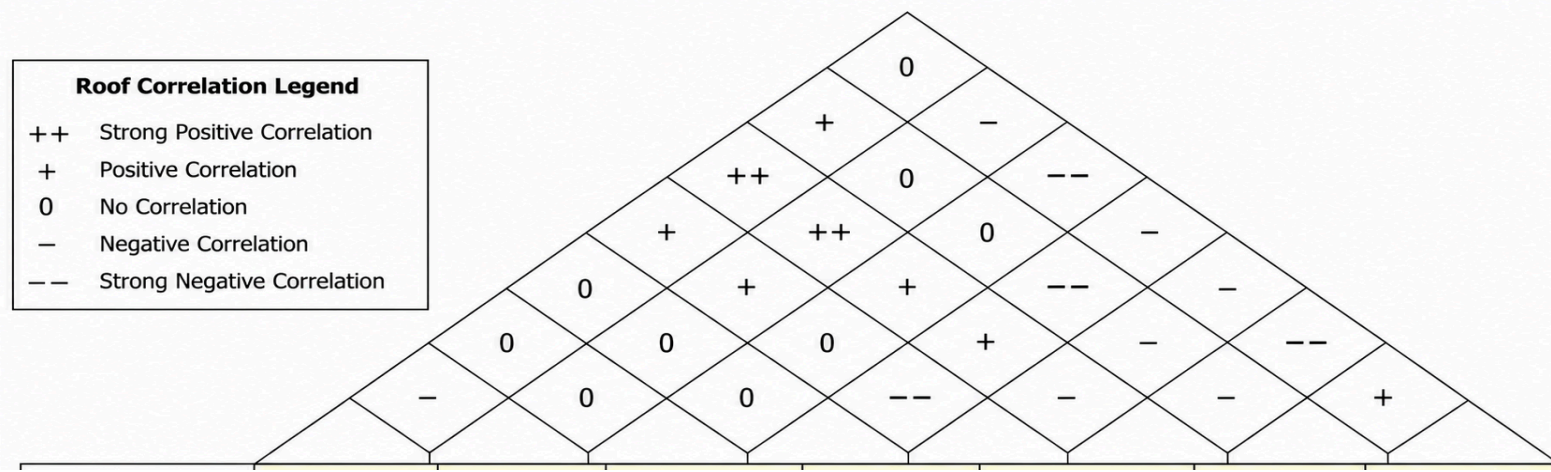


Table 15: House of Quality Matrix Level #2 (repeated for clarity)

	Part Characteristics	PC1: Shin Support Stiffness [Pa]	PC2: BOA Tension [N]	PC3: Resistance Fiber Loading Stiffness [N/m]	PC4: Outer Shell Thickness [mm]	PC5: Compression Interface Pressure [Pa]	PC6: Water Retention for Cooling [g]	PC7: Ventilation Hole Area Fraction [% open area]	PC8: Thermal Conductivity of Gel [W/m·K]
	Target Value	Resist 150 N·m Inversion	40-80 N	1700 N/m	2-4 mm	20-30 mmHg	≥ 18g	~20%	≥ 0.3 W/m·K
Engineering Characteristics	Weight from HOQ Level #1	Matrix Values							
EC1: Brace Security [Pa]	21.06%	9	3	3	3	3	0	1	3
EC2: ROM Control [N·m / °]	17.02%	9	9	9	3	1	0	0	0
EC3: Compression Level [Pa]	16.38%	0	9	1	3	9	0	1	3
EC4: Cooling Level [°C]	15.96%	0	0	0	0	0	9	9	9
EC5: Mass [g]	14.89%	3	1	1	9	1	9	3	3

EC6: Rehabilita- tion Loading [N·m]	19.5%	0	0	9	1	3	0	0	0	
	Absolute Weight		3.874	4.068	4.231	3.169	3.010	2.777	2.258	3.006
	Relative Weight		14.84%	14.49%	16.21%	12.14%	11.53%	10.64%	8.65%	11.51%
	Rank Order		3	2	1	4	5	7	8	6

Table 16 depicts the Bill of Materials (BOM) for a LoadStep prototype. Every part has characteristics that it satisfies.

Table 16: Bill of Materials (BOM) for a prototype of LoadStep

Item #	Component	Part	Quantity	Unit Cost	Total Cost	Purpose	Part Characteristics Satisfied
1	Shin Splint Structural Shell	PLA for prototype, injection molded later on	2	0.15kg x \$20 = \$3	\$6	Rigid support that stabilizes the ankle joint and prevents excessive movement.	PC1: Shin Support Stiffness
2	Carbon Fiber Skeleton Supports	Carbon fiber strips	16 8” Strips	\$1.12	\$17.93	Lightweight semi-flexible structural frame that provides high strength with minimal weight. Satisfies	PC4: Outer Shell thickness; PC7: Ventilation Hole Area Fraction
3	BOA Tensioning System	Off-the-shelf BOA dial and cable kit	1	\$15.99	\$15.99	Cable-based tightening system that evenly tightens the brace around the ankle	PC2: BOA Tension
4	Velcro Straps	Velcro straps	1 6” strip	\$2.25	\$2.25	External strap across the top of the bra that locks it in place during movement.	PC5: Compression Interface Pressure
5	Resistance Loading Fibers	Elastic resistance bands	2 1” x 6” x 0.25”	\$1.39	\$2.77	Elastic movements used for movement resistance and	PC3: Resistance Fiber Loading Stiffness

			strips			rehabilitation training	
6	Gel Cushion	Thermally conductive gel insert	1 8x6" sheet	\$1.24	\$1.24	Soft, cushioning layer placed between the brace and the skin of the ankle to cool and be comfortable	PC5: Compression Interface Pressure; PC8: Thermal Conductivity of Gel
7	Water Absorbing Inner Layer	Moisture absorbing cotton fabric	11.8x8in sheet	\$2.10	\$2.10	Flexible, water-absorbing fabric to keep the ankle cool	PC6: Water retention for cooling
8	Water Resistant Outer Layer	Nylon coated fabric	1 12" long piece	\$2.80	\$2.80	Prevents water from leaking out of the system	PC6: Water retention for cooling
8	Interior comfort Padding	EVA Foam	1 12" long piece	\$2.40	\$2.40	Increases comfort for user	PC5: Compression Interface Pressure
9	Cloth sleeve	Stretch compression fabric	1 12" long piece	\$3.67	\$3.67	Comfort interface and structural foundation for attachments	PC5: Compression Interface Pressure
10	Fasteners	1/4-20 Chicago Screws	10	\$10.42	\$10.42	Keep components attached to the brace	NA, fasteners just keep all components attached

Based on Table 16, the total cost for LoadStep is **\$61.57**.

3.4 Bass Model Forecast (EW)

Based on the competitor analysis table above (Table X), Product B, The Ankle Support Brace, is the most comparable product to LoadStep. Product B has good compression, is lightweight, and is relatively comfortable. To users, Product B might feel like it is helping their ankle heal, but the compression only helps mask pain, not actually heal it. Not only will our product integrate cooling, in addition to compression, to mask pain, but it will also incorporate active loading to help the ankle heal from the sprain. Even though the LoadStep will accelerate healing, it'll take some time to convince our target audience that it is better than their current ankle brace: products like Product B.

The Bass Model describes how new products get adopted in a population over time.³ The Bass Model is governed by the equation:

$$s(t) = m \frac{(q+p)^2}{p} \frac{e^{-(p+q)t}}{(1 + \frac{q}{p} e^{-(p+q)t})^2}$$

In this equation, m = the potential market, p = the coefficient of innovation, q = the coefficient of imitation, and $s(t)$ = sales at time “ t ”.

In the Bass Model, “ p ” is independent of prior adaptations. The higher the “ p ”, the stronger the external influence (e.g., advertising or marketing).¹³ The typical range for p is 0.01-0.03.¹³ We’ve set $p = 0.012$ for the LoadStep since there will be some initial external influence from athletic trainers, physical therapists, sports medicine clinics, and targeted rehabilitation product marketing, who are on board with our product, despite us being a small company. They will help introduce users to the device’s novel rehabilitation-loading functionality.

Also, in the Bass Model, “ q ” depends on the number of prior adaptations. The higher the value of “ q ”, the stronger the internal influence (e.g., word of mouth and social interactions).³ The typical range for q is 0.3-0.5.¹³ We’ve set $q = 0.32$ because adoption of rehabilitation and athletic recovery devices is heavily influenced by athlete recommendations, physical therapist referrals, trainer feedback, and observed recovery outcomes from peers. After the initial market introduction, word-of-mouth diffusion may start small because we are a small company; however, over time, it will increase significantly.

Both “ p ” and “ q ” are low for their ranges, given our startup status. It will take time to spread the word and get our target audience on board with our product.

Lastly, for the Bass Model, we will set $m = 1.39$ million users, representing our target market of individuals with ankle-related injuries (Section 3.1).

Plugging these values into the Bass Model equation and plotting the result in MATLAB yields an adoption curve (Figure 14).

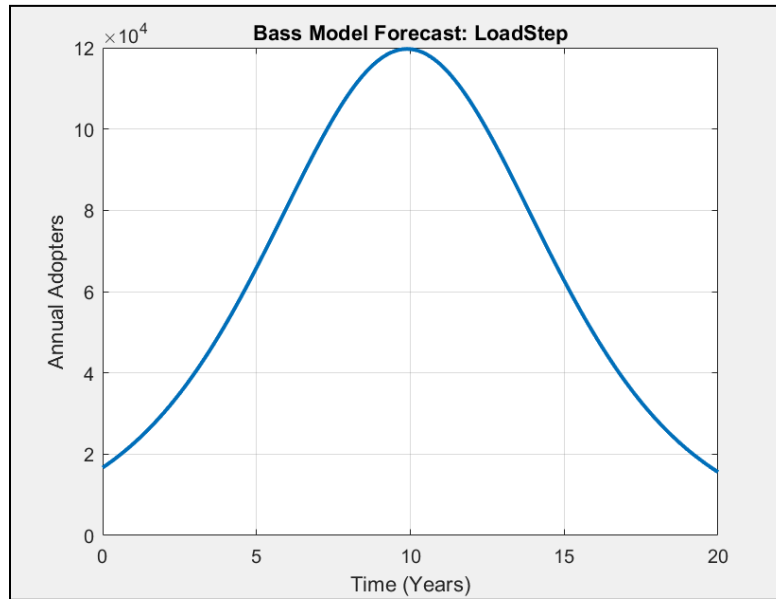


Figure 14: Bass Model Curve for LoadStep

The Bass Model demonstrates slow initial adoption, followed by growth, and eventually market saturation. Early adoption is driven by external influences (p), like athletic programs and clinician

recommendations. As more athletes, trainers, and physicians recommend LoadStep to others, adoption will increase rapidly through word-of-mouth (q). Peak adoption occurs at ~10 years. After this point, adoption declines since the market saturates. Overall, the curve suggests long-term adaptation for LoadStep, or at least until other companies jump on the bandwagon and also start producing ankle braces with resistance fibers for loading. Our patent will prevent this from occurring for at least 20 years, when the utility patent expires, which essentially corresponds with the time of market saturation according to the adoption curve.

Market Penetration is the percentage of the total target market that has adopted or purchased a product over a given period of time. In the Bass Model, market penetration is governed by the equation³:

$$F(t) = \frac{1 - e^{-(p+q)t}}{1 + \frac{q}{p}e^{-(p+q)t}}$$

Plugging this equation into MATLAB, along with the values for p and q, generates a curve (Figure 15).

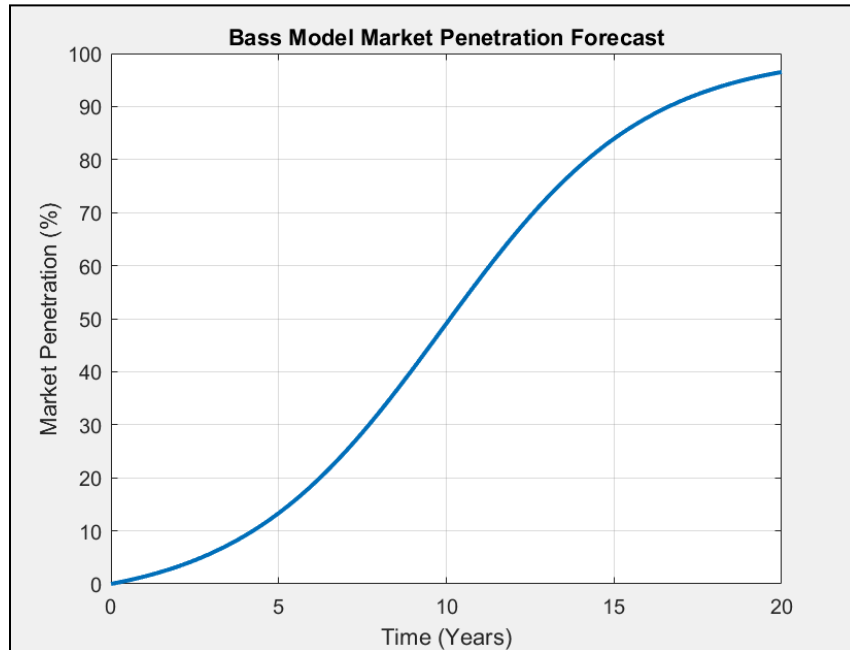


Figure 15: Market Penetration Forecast Curve

After 1 year, market penetration is about 1.4%, which is typical for a startup. After about 5 years, market penetration is around 13.5%. After around 10 years, market penetration is already at 50%, more than doubling from the value at 5 years. At 20 years, the market nearly saturates. This curve shows that market penetration for LoadStep is not linear but sigmoidal, with a slow start, a rapid increase in the middle, and leveling off as the market saturates.

4. Financial Requirements

4.1 Financial Executive Summary (CL)

Product Price: **\$89.99** (MSRP)

Expected Sales Volume

Sales volume is projected using the Bass Diffusion Model, calibrated to a total addressable market of approximately 1.39 million potential users with an innovation coefficient of $p = 0.012$ and an imitation coefficient of $q = 0.32$. Under this model, LoadStep is expected to reach approximately 19,460 unit sales in Year 1, growing to 82,010 cumulative units by the end of Year 3. By Year 10, cumulative sales are projected to reach approximately 695,000 units at peak market penetration. Annual unit sales for the first five years are detailed in Table 17 below.

Production Cost and Gross Margin

At prototype scale, the bill-of-materials cost is approximately \$61.57 per unit, yielding a gross margin of 31.6% at the \$89.99 MSRP, equivalent to a per-unit contribution of \$28.42 toward fixed costs and operating profit. Tooling costs are estimated at \$40,000 and treated as a one-time Year 0 capital cost in the NPV model below, with no recurring tooling expenditure anticipated within the five-year forecast horizon, given the projected production volumes.

Annual net contribution is calculated as $(\text{Price} - \text{COGS}) \times \text{Annual Unit Sales}$. Each year's net is discounted to present value using $PV = \text{Net} \div (1 + \text{WACC})^{\text{Year}}$, compounded annually at 5%. The \$40,000 tooling cost is deducted at Year 0 before discounting begins, and cumulative NPV tracks the running total of all discounted cash flows net of that initial investment

Table 17: 5-Year NPV Projection (WACC = 5%)

Years	Market Penetration	Cumulative Units	Annual Sales	Annual Net (\$)	Discount Factor	Present Value (\$)	Cumulative NPV (\$)
0	-	-	-	(\$40,000)	1	(\$45,000)	(\$40,000)
1	1.4%	19,460	19,460	0.553M	1.05	0.527M	0.487,
2	3.2%	44,480	25,020	0.71M	1.1025	0.645M	1.132M
3	5.9%	82,010	37,530	1.067M	1.1576	0.921M	2.053M
4	9.3%	129,270	47,260	1.343M	1.2155	1.105M	3.158M
5	13.5%	187,650	58,380	1.659M	1.2763	1.3M	4.458M

The \$40,000 tooling investment is recouped in Year 1, at which point the cumulative NPV turns positive. By Year 5, cumulative NPV reaches approximately \$4.458M, driven by annual net contribution growing from \$0.553M in Year 1 to \$1.659M in Year 5 as adoption accelerates.

4.2 Pricing Policy (CL)

Pricing Strategy

LoadStep is priced at \$89.99 MSRP. This price reflects the multi-component BOM cost of approximately \$61.57 at prototype scale and addresses consumer price sensitivity identified through conjoint analysis. Although consumers preferred lower prices, current material and manufacturing costs make a retail price below \$50 financially unfeasible without compromising essential rehabilitation functionality and product performance.

Value-Based Pricing

Conjoint analysis showed that consumers value affordability, comfort, ankle protection, and rehabilitation performance. Respondents indicated an 87.18% preference for adjustable progressive rehabilitation loading over fixed or no loading, and a 67.34% preference for a gel-padded interior with enhanced comfort and cooling. These features are LoadStep's core differentiators and align with the product configuration most preferred during testing.

Competitive Positioning

At \$89.99, LoadStep occupies a unique position in the ankle brace market. Basic braces, such as Product B (3.3), cost \$20-\$40 and provide only compression. In contrast, clinical-grade devices exceed \$200 and are typically prescribed rather than purchased directly. LoadStep bridges this gap by offering a device that actively accelerates recovery, not just pain management, while remaining affordable for college athletes who do not have insurance reimbursement.

No competitor currently offers active rehabilitation loading, ROM control, compression, and evaporative cooling in a single device at any price, especially under \$100. This combination justifies LoadStep's premium over basic braces. Our utility patent protects the rehabilitation-loading mechanism for up to 20 years, preventing larger manufacturers from replicating the design and undercutting our price. Distribution through physical therapists, athletic trainers, and sports medicine clinics will further support adoption, as clinician recommendations focus on recovery outcomes rather than price, which is where LoadStep excels.

4.3 Production Costs (GS/CL)

Manufacturing Process Selection

The brace is intended for low to medium batch production (custom rehabilitation devices rather than mass consumer goods). This favors cost-effective processes at small volumes.

The manufacturing processes for specific components of LoadStep are as follows:

- **Shin Splint Brace:** Cut to size, stitched into a shape cloth sleeve. A PLA plastic shin plate is thermoformed and fitted inside it. This combines the lightweight, rigid support of the plastic with the comfort and flexibility of the fabric.
- **Carbon Fiber Skeleton:** Manufactured using carbon fiber layup (vacuum bagging or compression molding) followed by curing in a mold. This process is chosen because it produces a high-strength-to-weight-ratio structure with precise control over stiffness and geometry. CNC trimming may be used post-curing for accuracy, but it depends on the extra cost.
- **Gel Cushion:** Produced using silicone or viscoelastic polymer casting/molding. This allows the gel to conform to the complex anatomical shape of the space around someone's foot while maintaining consistent cushioning and compression behavior. Mold-based production ensures repeatability and comfort-focused surface quality.
- **Over-top Velcro Strap:** Produced using textile cutting, stitching, and heat sealing.
- **Outer Cloth Lining:** Woven or knitted textile fabrication. Breathable synthetic fabrics with strong water absorption are selected to improve comfort and cooling duration.
- **Rehabilitative Loading Fibers:** Cut-to-size latex rubber or thermoplastic elastomers with stiffness “k” capable of withstanding the ankle torque requirements calculated in Section 2.11.

Tooling Costs

Tooling costs for LoadStep are estimated at \$40,000, covering three components that require custom molds. The PLA shin plate uses a single-sided thermoforming tool, estimated at \$4,000-\$7,000. Thermoforming is less capital-intensive than injection molding and suits LoadStep's low- to medium-volume batch production. The silicone gel cushion requires a precision compression mold to consistently reproduce the heel and arch geometry, estimated at \$5,000-\$12,000. The carbon fiber skeleton has the highest tooling cost, requiring a two-part machined aluminum or steel compression mold, estimated at \$15,000-\$40,000 depending on geometry complexity and the inclusion of CNC post-trimming fixtures. Textile components, including the cloth sleeve, over-top strap, and outer lining, do not incur tooling costs as they are cut and stitched using standard equipment at contract manufacturers. The \$40,000 is a one-time upfront investment; no recurring tooling costs are expected within the five-year forecast based on projected production volumes. All figures are approximate and based on industry benchmarks from sources including CustomPartNet and domestic short-run manufacturers.

4.4 Capital Investment (GS)

- **Equipment purchases:** Initial equipment would focus on enabling rapid prototype fabrication and testing with different-sized people to identify which physical changes are most impactful before scaling. This includes basic composite shaping tools (for carbon fiber layups or molds), cutting tools for straps and textiles, and assembly equipment such as heat presses or adhesives. The goal is to minimize upfront capital by prioritizing multi-use prototyping tools rather than full-scale manufacturing machinery.

- **Outsourcing options:** Specialized components can be outsourced to reduce cost and improve quality. The BOA tensioning system and carbon fiber frame are especially beneficial because of their manufacturing precision requirements and versatility in other, more established products. Gel padding and textile components can also be sourced from established suppliers that specialize in footwear. Outsourcing allows the design team to focus on optimization and testing rather than complex/expensive material production.
- **Funding strategy:** Funding can be secured through a combination of academic prototyping budgets, engineering grants, startup incubators, or crowdfunding if moving toward commercialization. Early-stage development would rely on low-cost prototyping funds, while later stages could seek sponsorship from sports medicine companies or rehabilitation clinics interested in wearable support technology.
- **Scaling workforce:** At the prototype stage, a small multidisciplinary team is sufficient, typically including mechanical design, materials expertise, and basic electronics support. As production scales, the workforce would expand to include manufacturing technicians, quality control specialists, and supply chain/logistics personnel. Outsourced manufacturing reduces the need for large in-house labor early on, allowing gradual scaling aligned with demand. Eventually, the team would need to run clinical trials to demonstrate that the technology improves recovery and strength. This would solidify Loadstep as the primary competitive ankle brace on the market, which doctors would recommend to patients.

5. Legal Requirements

5.1 Manufacturing Regulations (EW)

The LoadStep must meet several safety and regulatory requirements given its intended use as an orthopedic rehabilitation device for ankle sprains. The brace materials must be non-toxic, durable, and radiation-resistant during extended wear. The brace must withstand multiple loading cycles without mechanical failure. This is especially true of the resistance fiber loading system, which should ensure they don't loosen over time and no longer meet their goal of providing rehabilitative loading to the ankle. Compression limits must be within safe limits to avoid restricting circulation or causing discomfort during use. Since the device incorporates evaporative cooling and thermally conductive materials, additional precautions should also be taken to avoid excessive skin cooling and bacterial buildup during repeated use. Water-absorbing and cooling components should therefore be washable, durable, and resistant to leakage or material degradation over time. The device will likely have to be washed by hand to ensure none of the components loosen.

Since the LoadStep is a medical device, it is regulated by the FDA. The FDA classifies devices as either Class 1, Class 2, or Class 3, with Class 1 being the safest.¹⁴ The LoadStep would fall under Class 1 since it is a low-risk device, and other ankle braces are also classified as Class 1.¹⁵ The FDA characterizes Class 1 devices by their “minimal contact with patients and low impact on a patient’s overall health.”¹⁴ This means that the device is subject to only general controls, such as misbranding,

device registration, record-keeping, and good manufacturing practices.¹⁴ Even though the LoadStep includes active rehabilitation loading, the loading system is still passive and mechanically controlled by the user rather than electronically powered or automated. Therefore, the device is still low risk and comparable to competitor rehabilitation braces.

Environmental regulations are also crucial to consider. LoadStep is a reusable product with modular components that reduce waste. The user can keep the same brace but adjust the stiffness of the resistance fibers to toggle the amount of load they want. They can tighten the velcro strap to conform to their foot, allowing one brace to work for a range of shoe sizes. Manufacturing processes for components should aim to minimize plastic waste and failed prototype iterations. Many components are purchased off the shelf, including the BOA system, Velcro, laces, a water-absorbing layer, a water-resistant layer, and a thermally conductive gel. The components just need to be cut to size and will be ordered in quantities and sizes to minimize waste. The shin split portions will be additively manufactured from PLA for prototypes, which is more environmentally and cost-effective at small scales. Once the device is being mass-manufactured, it will be injection-molded, a low-waste process for large-scale production, especially since the molds can be used for multiple prototypes. Manufacturing processes should also consider environmental management standards such as ISO 14001, which focuses on reducing waste and improving sustainability throughout manufacturing operations.¹⁶

5.2 Product Use Regulations (EW)

Since our device includes rehabilitation loading, user safety compliance is very important to ensure the device functions correctly during use. The brace must fit securely without restricting circulation. Users will be advised on the proper degree of compression to ensure a proper fit. Instructions on how to use the device will clearly indicate what each component is used for (e.g., the BOA system for ROM restriction, laces to adjust compression, and resistance fibers for active loading). The cooling chambers will also have specific protocols to ensure no bacterial buildup in the water-absorbing layers, which could lead to infection. The user will be advised to wash the device at least weekly, or after extended athletic use, to prevent this from occurring.

Additionally, it is highly recommended that the LoadStep be used under the guidance of athletic trainers, physical therapists, or clinicians to ensure safe rehabilitation progression and proper brace adjustment throughout recovery. These individuals are qualified to tell users how much compression, ROM, and loading the user should set the device to. If unable to receive advice from a physician or trainer, the user should set the system to restrict ROM and provide the least amount of loading. They should then loosen the ROM restrictions and increase the loading based on what they feel comfortable with. This will ensure the user does not accidentally twist their ankle in an unwanted direction due to too much ROM or load the ankle too much that it is no longer rehabilitative.

5.3 Environmental Impact - During Production (CL)

During Production

Material Use

LoadStep minimizes environmental impact by prioritizing careful material selection. Most components, including the BOA system, Velcro straps, water-absorbing cotton, water-resistant nylon, and thermally conductive gel, are sourced off-the-shelf and cut to size, reducing waste compared to custom manufacturing. Carbon fiber strips are purchased in pre-cut lengths to further minimize offcuts.

For prototyping, additive manufacturing produces the shin splint structural shell from PLA (polylactic acid). PLA provides an optimal solution for this application because it is a “non-toxic, biodegradable, compostable, biobased polymer,” and it generates minimal hazardous waste during printing.¹⁷ Disposal is specified via industrial composting facilities, recognizing that composting requires sustained temperatures above 58°C and is not suitable for home composting streams.

At the production scale, injection molding significantly reduces per-unit material waste. Injection molding is “a naturally low waste process... each cycle uses just the amount of plastic needed to fill the mold,” and relative to subtractive methods like CNC machining, it produces relatively low scrap rates compared to traditional manufacturing processes.¹⁸ Any thermoplastic offcuts generated are further recovered: “post-industrial residue can be broken down and returned to the injection molding machine, or plastic pieces can be melted into pellets for reuse.”¹⁹

Waste Reduction

Several design decisions directly aim to reduce manufacturing waste. Sourcing off-the-shelf components eliminates the need for complex custom fabrication and reduces waste from failed development iterations. The modular architecture ensures that individual components, such as resistance bands, gel inserts, and straps, are easily replaced, thereby extending the lifespan of the structural frame and minimizing the need for full device disposal.

Carbon fiber manufacturing presents a known tradeoff: “Approximately 40% of CFRP waste is generated during the manufacturing stage.”²⁰ This challenge is addressed by ordering carbon fiber components in optimized batch quantities to minimize unused material and by prioritizing suppliers certified to ISO 14001.¹⁶ This approach also supports partnerships with sports medicine companies and institutional buyers who increasingly require environmental compliance from suppliers.

5.4 Environmental Impact - End of Life (CL)

Recycling

LoadStep leverages a modular architecture to maximize repairability and minimize unnecessary full product disposal at the end of its lifespan. Key components, such as the thermoplastic shell, metal

fasteners, textile fabrics, and certain plastic support structures, are designed for disassembly and can be recycled where local facilities are available. The BOA system and fastening hardware are engineered for independent reuse or refurbishment, maximizing resource efficiency and minimizing waste from the main brace assembly.

The modular design enables precise replacement of high-wear components, such as resistance bands, gel inserts, straps, and fabric liners. This targeted maintenance extends the product's lifespan, reduces long-term material waste, and provides a superior alternative to traditional single-structure ankle braces, which often require full replacement if any single component fails.

Disposal

Some LoadStep materials remain incompatible with standard municipal recycling. Carbon fiber reinforcement strips, thermally conductive gel inserts, EVA foam padding, and elastic rehabilitation bands require specialized disposal due to their advanced composite or polymer construction. Carbon fiber is particularly challenging to manage at the end of life: “since carbon fiber composites are not biodegradable, their disposal poses environmental challenges.”²¹ Unlike thermoplastics, carbon fiber cannot be remelted or reformed, limiting recyclability and necessitating industry-specific processing.

Despite these material limitations, LoadStep is engineered for repeated, long-term use rather than short-term disposal. The design facilitates the repair and replacement of modules, significantly reducing the need for full product disposal and supporting sustainability objectives. Manufacturing and material choices maximize durability, extend service life, and minimize environmental impact throughout the product lifecycle.

5.5 Liability Issues (GS)

Misuse Cases

Incorrect use includes wearing the brace on the wrong limb or in an incorrect orientation, using it without proper sizing adjustments, or relying on it as a complete replacement for medical treatment or rehabilitation guidance. Another incorrect use is the use of non-approved resistance levels or configurations during rehabilitation exercises, which could place abnormal loads on the ankle joint and increase the risk of injury.

Taken together, the Loadstep *could* create liability issues if it doesn't provide sufficient support or is used beyond its design limits. There's also a risk if it's treated or marketed as a medical-grade device without the proper certification, since that could lead to regulatory problems. On top of that, if key parts like the carbon fiber frame or straps fail during use, it could cause injury by suddenly reducing support or restricting movement in an unsafe way. Our team would need to ensure that we never step over any legal lines until certification is achieved that our product can reliably perform in specific cases that need to be identified.

5.6 Failure Mode and Effects Analysis (FMEA) (GS)

Failure modes + RPN + Risk Mitigation

- Carbon fiber “skeleton” cracking or fracturing under high load
 - Severity: 2
 - Occurrence: 3
 - Detectability: 2
 - **RPN Rating: 12**
 - RPN Reduction Strategy: Use higher-grade carbon fiber, increase the safety factor, and optimize the design to better carry the load.
- BOA tensioning system loosening or cable snapping
 - Severity: 6
 - Occurrence: 1
 - Detectability: 3
 - **RPN Rating: 18**
 - RPN Reduction Strategy: Reinforce cable routing paths, include regular inspection/replacement intervals for cables and dial mechanisms.
- Strap/Velcro fatigue leading to loss of secure fit
 - Severity: 4
 - Occurrence: 5
 - Detectability: 2
 - **RPN Rating: 40**
 - RPN Reduction Strategy: Select high-cycle durability Velcro, increase strap width to reduce localized stress, and provide more secure contact.
- Gel cushioning compression set (loses thickness over time)
 - Severity: 2
 - Occurrence: 2
 - Detectability: 2
 - **RPN Rating: 8**
 - RPN Reduction Strategy: Use high-resilience viscoelastic gel materials.
- Resistance bands overstretching or snapping (very unlikely)
 - Severity: 3
 - Occurrence: 1
 - Detectability: 1
 - **RPN Rating: 3**
 - RPN Reduction Strategy: Specify elastic limits well below failure strain in an instruction manual (along with other user guidelines to prevent misuse).
- Improper fit causing uneven compression and pressure points
 - Severity: 4
 - Occurrence: 6
 - Detectability: 2
 - **RPN Rating: 48**

- RPN Reduction Strategy: Add adjustable fitting zones for the BOA and straps, include sizing guidelines, and incorporate pressure-distribution padding to reduce localized stress concentrations.
- Improper installation causes further tissue damage with overuse
 - Severity: 7
 - Occurrence: 6
 - Detectability: 5
 - **RPN Rating: 210**
 - RPN Reduction Strategy: Provide clear fitting instructions with visual guides and alignment markers, and use warnings or optional pressure feedback to prevent over-tightening and misuse.

5.7 Industry Standards (GS)

LoadStep is designed with consideration for industry standards relevant to orthopedic rehabilitation devices and wearable support systems. Materials in contact with the skin should comply with the International Organization for Standardization ISO 10993 biocompatibility standards to ensure safe long-term wear.²² Manufacturing and quality control processes would aim to follow ISO 13485 standards for medical device quality management²³, while environmental practices would align with ISO 14001 standards for waste reduction and sustainable manufacturing.¹⁶

Since LoadStep is a mechanical device with no powered electronics, UL electrical safety standards are not currently applicable.

5.8 Intellectual Property (EW)

A utility product describes “how an article works,” while a design patent describes “how an article looks.”²⁴ Our team plans to pursue a utility patent for LoadStep because the novelty of the product primarily lies in the integration or rehabilitation of loading fibers in an ankle brace. The device adds this component and combines other features found in various products on the market, such as compression and cooling, into one device. We may pursue a design patent, but it would only protect the product's aesthetic. Our novelty lies more in function than in appearance, making a utility patent ultimately more crucial for protecting our design.

Trade secrets are intellectual property rights on commercially valuable, confidential information.²⁵ We will have trade secrets for the exact resistance values for the loading fibers and why they are considered rehabilitative loading rather than harmful. We will also have trade secrets for the exact material makeup of our product and each material’s mechanical and thermal properties. This will ensure that no one can copy our cooling mechanism or create a product that stays cool for as long as ours.

The product name, LoadStep, will be trademarked to ensure no one else can create a product with the same name as ours. We want our product to stand out to potential customers. If another product had the same name, our product could be hidden in the market. Other trademark opportunities could

include our product slogan, “Recover Through Motion,” to further emphasize how our product allows users to go about their everyday lives while still healing their ankle sprain.

Figure 16 displays our product logo. This logo emphasizes our product’s goal of facilitating healing while still allowing the user to move. The arrows illustrate how this goal is achieved by increasing the rehabilitative load on the ankle.



Figure 16: LoadStep Logo

5.9 Next Steps (EW)

A major next step for LoadStep would be to begin to prototype the product. This will allow us to verify that our cooling, compression, and BOA tensioning mechanisms work as expected. Resistance fibers will first undergo mechanical testing under physiologically relevant loading conditions to evaluate their fatigue performance. Then, efficacy testing will likely involve IRB-approved clinical studies to determine whether progressive rehabilitation loading measurably improves healing outcomes compared to traditional ankle braces. These studies could evaluate metrics such as recovery time, pain levels, ankle stability, range of motion, and return-to-play timelines.

Following successful prototype and verification/validation testing, a utility patent would be filed to protect the product's functional mechanism. This would prevent other companies from creating a device that integrates rehabilitative loading, ROM control, compression, and cooling to deliver a comfortable, supportive ankle brace that truly accelerates healing, not just masks pain. Then, the product could be licensed with orthopedic rehabilitation or sports medicine companies interested in our novel approach to improving ankle sprain healing while still allowing players to go about their day-to-day lives.

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